Geometry Review: Unit 5 — Right Triangles & Trigonometry

Answer Key

Station 1: The Pythagorean Theorem

1. 10
2. 13
3. 15
4. $\sqrt{65} \approx 8.1$
5. $2\sqrt{14} \approx 7.5$ $\quad \sqrt{81-25} = \sqrt{56}$
6. Yes; $9^2 + 12^2 = 81 + 144 = 225 = 15^2$
7. Obtuse $\quad 7^2 + 8^2 = 113 < 144 = 12^2$
8. 12 ft
9. 15 in
10. $7\sqrt{2} \approx 9.9$

Station 2: Special Right Triangles: 45°-45°-90° and 30°-60°-90°

1. $7\sqrt{2}$
2. $6\sqrt{2}$ $\quad \dfrac{12}{\sqrt{2}} = \dfrac{12\sqrt{2}}{2}$
3. $5\sqrt{2}$
4. $5\sqrt{3};\ 10$
5. $9;\ 9\sqrt{3}$
6. $6;\ 12$
7. $3\sqrt{3};\ 6\sqrt{3}$ $\quad \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$
8. 8
9. $5\sqrt{3}$
10. $16\sqrt{3}$ $\quad \frac{1}{2}(8)(4\sqrt{3})$

Station 3: Trigonometric Ratios: Sine, Cosine, Tangent

1. $\dfrac{8}{17},\ \dfrac{15}{17},\ \dfrac{8}{15}$
2. $\dfrac{12}{13},\ \dfrac{5}{13},\ \dfrac{12}{5}$
3. $\dfrac{7}{25},\ \dfrac{24}{25},\ \dfrac{7}{24}$
4. $\cos\theta = \dfrac{12}{13},\ \tan\theta = \dfrac{5}{12}$
5. $\sin\theta = \dfrac{4}{5},\ \tan\theta = \dfrac{4}{3}$
6. $70°$ $\quad$ *sine and cosine of complementary angles are equal*
7. $\tan\theta = \dfrac{3}{4};\ \text{hyp} = 10;\ \sin\theta = \dfrac{3}{5}$
8. $\dfrac{1}{2},\ \dfrac{1}{2},\ 1,\ \sqrt{3}$
9. $15;\ 9$
10. $\left(\dfrac{3}{5}\right)^2 + \left(\dfrac{4}{5}\right)^2 = \dfrac{9}{25} + \dfrac{16}{25} = 1$

Station 4: Solving Right Triangles with Trigonometry

1. $x = 12\sin 35° \approx 6.9$
2. $x = \dfrac{9}{\cos 52°} \approx 14.6$
3. $x = 15\tan 28° \approx 8.0$
4. $x = \dfrac{20}{\tan 63°} \approx 10.2$
5. $\theta = \sin^{-1}\left(\dfrac{7}{10}\right) \approx 44.4°$
6. $\theta = \cos^{-1}\left(\dfrac{5}{13}\right) \approx 67.4°$
7. $\theta = \tan^{-1}\left(\dfrac{9}{4}\right) \approx 66.0°$
8. $50\tan 40° \approx 42.0$ m
9. $20\sin 70° \approx 18.8$ ft
10. $\tan^{-1}\left(\dfrac{120}{200}\right) \approx 31.0°$ $\quad$ *angle of depression = angle of elevation from the boat*